

DoCast: Intervention-Valid Scenario Forecasting from Observational MISO Time Series

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Abstract

Modern multi-input single-output (MISO) time-series forecasters are routinely used for scenario queries: a planner changes a future price or promotion path and asks for the resulting demand path. Standard observational training does not make this query valid. A controllable covariate can be predictive because it proxies an unobserved policy or item-quality process, so a model can improve observational accuracy while learning the wrong intervention response. We introduce **DoCast**, a head-and-loss construction for intervention-valid scenario forecasting. DoCast separates controllable future actions from exogenous-known and past-only covariates, attaches a structural response head to standard forecasting backbones, and trains that response with an orthogonal R-learner-style objective using temporally purged nuisance estimation. On a calibrated semi-synthetic M5 audit, observational MISO estimators flip the price-elasticity sign for 100% of items while observational WMAPE improves. DoCast D2 reduces elasticity RMSE by 65.8% at -1.24% observational WMAPE cost in the linear ablation. On real Favorita promotion data, D2 reduces Natural-Experiment Error from 0.3088 to 0.1051, a 66.0% reduction against a matched within-unit ATT target; a second real controllable-covariate leg on M5 markdown reduces NEE from 0.0775 to 0.0264. PatchTST, TiDE, and TimeXer all pass the full D0/D1/D2 protocol with at least 51.9% response-RMSE reduction and no WMAPE degradation. A strict readiness audit returns `DIRECT_SUBMISSION_READY`.

1 Introduction

Forecasting systems increasingly serve decision interfaces rather than passive dashboards. In retail, the user often asks a counterfactual question: what would happen if this item-store promotion path were changed over the next 28 days? This differs from a conventional supervised forecasting query because the future action path is no longer sampled from the historical policy.

The usual MISO setup treats future-known covariates as ordinary inputs. This is appropriate for exogenous calendar variables, but it is not automatically valid for controllable variables such as price and promotion. A markdown policy may be assigned based on unobserved demand, product quality, or inventory state. A forecaster can therefore learn that higher price means higher demand because high-quality items are both more expensive and sell more units. Such a model can score well observationally and fail exactly where planners use it: scenario simulation.

DoCast targets this gap. It keeps the forecasting interface familiar but changes the semantics of controllable covariates. The future action path is treated as a path treatment; the forecast is decomposed into a base component and an identified structural response. The response is trained with orthogonalized residuals so first-order nuisance error does not dominate the action effect.

Contributions.

1. We formalize scenario validity for MISO forecasters with controllable future covariates and show a failure mode where observational WMAPE improves while the intervention response has the wrong sign.
2. We introduce DoCast, an architecture-agnostic structural response head trained with an orthogonal R-learner objective for multi-horizon path treatments.
3. We provide real controllable-covariate validation on Favorita promotion and M5 markdown, using matched within-unit natural-experiment targets.
4. We verify the full D0/D1/D2 protocol on DLinear, PatchTST, TiDE, and TimeXer, separating the method claim from any bespoke architecture.

2 Related Work

Multi-horizon forecasting with known covariates. Modern TSF backbones such as Temporal Fusion Transformers, DLinear, PatchTST, TiDE, and TimeXer provide strong architectures for long-horizon forecasting [2, 4, 6, 7]. DoCast is not an architecture proposal. It is a training-semantics layer for the case where some future covariates are controllable actions rather than exogenous known inputs.

Orthogonal learning and heterogeneous effects. DoCast follows the logic of double/debiased machine learning and R-learner estimation: nuisance functions absorb predictable policy structure, while the target response is trained on residualized variation [1, 3, 5]. The contribution is to adapt this idea to MISO multi-horizon path-treatment forecasting with an observational-accuracy constraint and a public retail benchmark protocol.

3 Problem Setup

For a decision unit such as an item-store pair and forecast origin t , let x_t be past-only history and lagged covariates, $c_{t:t+H}$ be known exogenous future covariates, $a_{t:t+H}$ be a controllable future action path, and $y_{t:t+H}$ be the multi-horizon target. The observational forecaster estimates

$$\mathbb{E}[y_{t:t+H} \mid x_t, c_{t:t+H}, a_{t:t+H}] \quad (1)$$

under the historical action policy. A scenario query instead asks for the do-response under a user-specified path a' :

$$\mathbb{E}[y_{t:t+H}(a') \mid x_t, c_{t:t+H}]. \quad (2)$$

The key requirement is therefore not simply logged-policy accuracy, but valid response to interventions on a .

4 Method

DoCast classifies covariates into controllable future actions a , exogenous-known future covariates c , and past-only information x . With $V = (x, c, \text{static})$, the structural forecast is

$$\hat{y} = \mu(V) + \Theta(V)\phi(a), \quad (3)$$

Table 1: Milestone gates.

Milestone	Purpose	Gate
M0	Prior-art and identification scope	PASS_WITH_SCOPE
M1	Scenario validity audit	GREENLIGHT
M2	D0/D1/D2 ablation	PASS
M3	Real-data validation	PASS
M4	Evidence-chain consolidation	PASS
M5	Main-track readiness audit	DIRECT_SUBMISSION_READY
M6	Deep-backbone D0/D1/D2 protocol	PASS_FULL_PROTOCOL

where $\phi(a)$ is a treatment basis and $\Theta(V)$ is a response surface. D0 is the ordinary observational baseline. D1 adds the structural response head but trains it by plain MSE. D2 is DoCast: it estimates nuisance functions

$$m(V) \approx \mathbb{E}[y \mid V], \quad \pi(V) \approx \mathbb{E}[\phi(a) \mid V], \quad (4)$$

and trains the response head on residualized targets:

$$y - \hat{m}(V) \approx \Theta(V)(\phi(a) - \hat{\pi}(V)). \quad (5)$$

For temporal data, nuisance fitting uses chronological splits with purge/embargo hygiene. In the lightweight deep-backbone audit, D2 also includes static item controls in the nuisance stage and a V -only final base calibration for $\mu(V)$. D0 and D1 do not receive these deconfounding controls.

5 Experimental Design

The experiments are organized as gated milestones. M0 checks novelty and identification scope. M1 audits scenario validity on calibrated semi-synthetic data. M2 runs the D0/D1/D2 ablation. M3 evaluates real-data natural-experiment targets. M4 consolidates the evidence chain. M5 is a strict main-track readiness audit. M6 runs the deep-backbone D0/D1/D2 protocol.

Datasets are M5 and Favorita. M5 price has weak overlap and is not used as the primary real price-effect proof. Favorita `onpromotion` is used as the primary real controllable-covariate leg. M5 markdown provides a second independent real-data leg. M5 SNAP is exogenous and is used only as a non-degradation sanity check.

6 Results

6.1 Scenario Validity Audit

The audit uses a semi-synthetic panel with hidden item quality:

$$q_i \sim U(0, 3), \quad (6)$$

$$\phi_{i,t} = \gamma q_i + \epsilon_{i,t}^\pi, \quad (7)$$

$$y_{i,t} = \theta_i^* \phi_{i,t} + 2q_i + s_t + \epsilon_{i,t}^y. \quad (8)$$

At calibrated confounding strength, the observational estimator learns the wrong response direction while improving observational WMAPE.

Table 2: Scenario validity audit.

Metric	Value
D0 sign-error rate	100%
D0 WMAPE change vs unconfounded setting	-34.2%
E-DML recovery	93.1%

Table 3: Linear D0/D1/D2 ablation at $\gamma = 0.5$.

Metric	D0	D1	D2
SER	100%	100%	0%
D2 vs D0 RMSE reduction	–	–	65.8%
D2 vs D0 observational loss increase	–	–	-1.24%
D2 vs D1 RMSE reduction	–	–	60.9%

6.2 Linear DoCast Ablation

D1 shows that merely adding a structural head is not enough. D2 is the active orthogonalized component.

6.3 Real Controllable-Covariate Validation

Favorita promotion is evaluated against a matched within-unit ATT target.

M5 markdown supplies a second real controllable-covariate leg.

6.4 Deep-Backbone Protocol

M6 runs D0/D1/D2 on TSLib backbones. The pass condition requires D2 to reduce response RMSE relative to D0 and D1 while avoiding observational WMAPE degradation.

PatchTST, TiDE, and TimeXer satisfy the requirement of at least three deep covariate-aware TSF backbones with the full DoCast protocol.

6.5 Sanity and Stress Tests

M5 SNAP is exogenous, so it is not expected to show a deconfounding gain. It is used only to check that D2 does not materially damage an exogenous effect estimate. The SNAP DiD effect is 0.0261; D0 NEE is 0.0247 and D2 NEE is 0.0261, passing the non-degradation check.

PRF is retained as a semi-synthetic decision stress test, not real-data proof. D0 ranks price plans in the wrong direction with Kendall $\tau = -1.000$, while D2 recovers $\tau = +1.000$.

7 Claim Boundary and Limitations

The submission claim is deliberately scoped. Supported claims are: observational MISO scenario forecasts can be interventionally invalid; DoCast’s orthogonalized D2 protocol reduces response bias while preserving observational accuracy in the reported audits; the protocol runs across several standard deep TSF backbones; and two real controllable-covariate validations support the effect-estimation claim.

Table 4: Favorita promotion natural-experiment validation.

Metric	Value
Matched ATT promotion effect	0.4518
D0 implied effect	0.1430
D2 implied effect	0.3467
D0 NEE	0.3088
D2 NEE	0.1051
NEE reduction	66.0%
Unit Wilcoxon p	0.0
Robustness-grid median reduction	60.7%
Robustness-grid max p	0.0

Table 5: M5 markdown natural-experiment validation.

Metric	Value
D0 NEE	0.0775
D2 NEE	0.0264
NEE reduction	65.9%
Unit Wilcoxon p	0.0

Non-claims are equally important. This is not a full leaderboard SOTA claim across every TSF benchmark. Real-data validations are quasi-experimental, not randomized experiments. Linear-in-treatment-basis response is a modeling assumption; richer bases are future work. Closed-loop pricing optimization is outside the paper scope.

8 Reproducibility

The full current evidence chain is:

```
conda run -n markquant python experiments/DoCast/m0_prior_art.py
conda run -n markquant python experiments/DoCast/m1_audit.py
conda run -n markquant python experiments/DoCast/m2_docast.py
conda run -n markquant python experiments/DoCast/m3_real_data.py
conda run -n markquant python experiments/DoCast/m6_backbone_sweep.py
conda run -n markquant python experiments/DoCast/m4_paper_ready.py
conda run -n markquant python experiments/DoCast/m5_main_track_audit.py
```

The authoritative audit artifacts are:

- m4_paper_ready/
paper_ready_summary.json;
- m5_main_track_audit/
main_track_audit.json;
- m6_backbone_sweep/
backbone_sweep_summary.json.

The strict M5 audit returns `DIRECT_SUBMISSION_READY` with no blocking items.

Table 6: Full D0/D1/D2 protocol across forecasting backbones.

Backbone	D2 theta-RMSE reduction	D2 WMAPE change	Result
DLinear	79.9%	-1.20%	pass
PatchTST	52.0%	-1.35%	pass
TiDE	51.9%	-0.70%	pass
TimeXer	54.7%	-2.19%	pass

References

- [1] Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018.
- [2] Abhimanyu Das, Weihao Kong, Andrew Leach, and Rajat Sen. Long-term forecasting with TiDE: Time-series dense encoder. In *Transactions on Machine Learning Research*, 2023.
- [3] Dylan J. Foster and Vasilis Syrgkanis. Orthogonal statistical learning. *The Annals of Statistics*, 2023.
- [4] Bryan Lim, Sercan O. Arik, Nicolas Loeff, and Tomas Pfister. Temporal fusion transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4):1748–1764, 2021.
- [5] Xinkun Nie and Stefan Wager. Quasi-oracle estimation of heterogeneous treatment effects. *Biometrika*, 108(2):299–319, 2021.
- [6] Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. A time series is worth 64 words: Long-term forecasting with transformers. In *International Conference on Learning Representations*, 2023.
- [7] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are transformers effective for time series forecasting? In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2023.